

THE $(\infty, 2)$ -FUNCTORIALITY OF THE SPAN CONSTRUCTION FROM ADEQUATE TRIPLES

ABSTRACT. We construct the span construction as an $(\infty, 2)$ -functor on ∞ -categories with pullbacks, pullback-preserving functors, and cartesian natural transformations. The span-specific input is entirely the theory of adequate triples of Haugseng–Hebestreit–Linskens–Nuiten; the general formalism is developed for arbitrary monoidal bases using Gepner–Haug seng categorical algebras and Heine’s functorial comparison from closed actions to enriched precategories. The underlying ∞ -functor preserves small limits. This manuscript and its mathematical argument were generated with ChatGPT (GPT-5.6 Sol) in dialogue with the user.

1. STATEMENT

We fix Grothendieck universes and enlarge them whenever necessary. We use the word *anima* (plural *animae*) for a homotopy type, i.e. an ∞ -groupoid. We use Gepner–Haug seng categorical algebras as a model for enriched ∞ -precategories, and their complete categorical algebras as a model for enriched ∞ -categories [GH15]. In particular, a complete Cat_∞ -enriched categorical algebra will be called an $(\infty, 2)$ -category.

Let $\text{Cat}_\infty^{\text{pb}}$ be the ∞ -category of ∞ -categories admitting pullbacks and pullback-preserving functors. A natural transformation $\alpha : F \Rightarrow G$ is *cartesian* if for every arrow $f : x \rightarrow y$ the naturality square

$$\begin{array}{ccc} F(x) & \xrightarrow{F(f)} & F(y) \\ \alpha_x \downarrow & & \downarrow \alpha_y \\ G(x) & \xrightarrow{G(f)} & G(y) \end{array}$$

is a pullback. For $\mathcal{C}, \mathcal{D} \in \text{Cat}_\infty^{\text{pb}}$, let

$$\text{Fun}_{\text{pb}}^{\text{cart}}(\mathcal{C}, \mathcal{D}) \subseteq \text{Fun}(\mathcal{C}, \mathcal{D})$$

be the subcategory whose objects are pullback-preserving functors and whose morphisms are cartesian natural transformations.

Theorem 1.1. *There exists a complete Cat_∞ -enriched category*

$$\mathbf{Cat}_\infty^{\text{pb, cart}}$$

with the following properties.

- (1) *Every object is equivalent to one represented by an ∞ -category $\mathcal{C}$ admitting pullbacks.*

(2) For represented objects $\mathcal{C}, \mathcal{D}$ there is a canonical equivalence

$$\mathbf{Cat}_\infty^{\text{pb}, \text{cart}}(\mathcal{C}, \mathcal{D}) \simeq \text{Fun}_{\text{pb}}^{\text{cart}}(\mathcal{C}, \mathcal{D}).$$

There is a $\mathbf{Cat}_\infty$ -enriched functor

$$\mathbf{Span}: \mathbf{Cat}_\infty^{\text{pb}, \text{cart}} \longrightarrow \mathbf{Cat}_\infty$$

whose restriction to represented objects, 1-morphisms, and 2-morphisms is respectively

$$\mathcal{C} \longmapsto \text{Span}(\mathcal{C}), \quad F \longmapsto \text{Span}(F), \quad \alpha \longmapsto \text{Span}(\alpha).$$

Here $\mathbf{Cat}_\infty$ denotes the Gepner–Haugsgeng completion of the categorical self-enrichment of $\mathbf{Cat}_\infty$.

Moreover, the ordinary ∞ -functor

$$\text{Cat}_\infty^{\text{pb}} \longrightarrow \mathbf{Cat}_\infty, \quad \mathcal{C} \longmapsto \text{Span}(\mathcal{C}),$$

preserves all small limits.

The first part supplies the coherent $(\infty, 2)$ -functoriality asked for in Harington–Mimram Conjecture 3.5 [HM24, Conjecture 3.5]. The final assertion concerns ordinary ∞ -categorical limits; see Remark 8.2.

2. CATEGORICAL ALGEBRAS AND COMPLETION

We assume the standard theory of ∞ -categories, including cartesian and cocartesian fibrations, straightening/unstraightening, and monoidal ∞ -categories. The less standard notions used below are recalled here in the form needed for the proof, with references to their original definitions.

2.1. Algebraic patterns and many-object associative algebras. We first translate the generalized non-symmetric operads of Gepner–Haugsgeng into the language of algebraic patterns. An *algebraic pattern* is an ∞ -category $\mathcal{O}$ equipped with an inert–active factorization system

$$(\mathcal{O}^{\text{int}}, \mathcal{O}^{\text{act}})$$

and a full subcategory $\mathcal{O}^{\text{el}} \subseteq \mathcal{O}^{\text{int}}$ of elementary objects; this is the terminology of Chu–Haugsgeng [CH21, Definition 2.1]. The simplex category gives two basic patterns. On Δ^{op} , inert maps are opposites of interval inclusions and active maps are opposites of endpoint-preserving maps. We write

$$\Delta^{\text{op}, \natural} \quad \text{for the pattern with elementary objects } [0], [1],$$

and

$$\Delta^{\text{op}, \flat} \quad \text{for the pattern with the single elementary object } [1].$$

This notation follows Barkan–Haugsgeng–Steinebrunner [BHS25, Example 3.1.4]; the same two patterns occur in Chu–Haugsgeng [CH21, Example 3.3]. Segal $\Delta^{\text{op}, \natural}$ -objects are category objects, while Segal $\Delta^{\text{op}, \flat}$ -objects are associative monoids.

The pattern-theoretic analogue of an operad is a fibrous pattern. For the present base, the comparison is exact: generalized non-symmetric ∞ -operads in the sense of Gepner–Haugsgaard are precisely weak Segal $\Delta^{\text{op}, \natural}$ -fibrations [CH21, Example 9.8(ii)], equivalently fibrous $\Delta^{\text{op}, \natural}$ -patterns [BHS25, Example 4.1.9]. Thus the active/inert structure used by Gepner–Haugsgaard can be read as the active/inert structure of the algebraic pattern $\Delta^{\text{op}, \natural}$, and a morphism of generalized non-symmetric ∞ -operads is a morphism over this base preserving the cocartesian lifts of inert maps; compare Gepner–Haugsgaard [GH15, Definition 2.4.2] with Barkan–Haugsgaard–Steinebrunner [BHS25, Definition 4.1.12].

For an anima T , Gepner–Haugsgaard define

$$\Delta_T^{\text{op}} \longrightarrow \Delta^{\text{op}}$$

as the cocartesian fibration classified by the simplicial ∞ -category

$$[n] \longmapsto T^{n+1}.$$

See Gepner–Haugsgaard [GH15, Remark 4.1.2 and Lemma 4.1.3]. In algebraic-pattern language this is a Segal $\Delta^{\text{op}, \natural}$ -fibration: the Segal decomposition over $[n]$ records a string of n arrows together with its $n+1$ vertices. In particular it is a generalized non-symmetric ∞ -operad.

Let

$$\mathcal{V}^{\otimes} \longrightarrow \Delta^{\text{op}}$$

be a non-symmetric monoidal ∞ -category. Equivalently, it is a Segal $\Delta^{\text{op}, \flat}$ -fibration [CH21, Example 9.2(ii)]; after forgetting the contractibility condition at $[0]$, it is also a generalized non-symmetric ∞ -operad. A T -object categorical algebra in $\mathcal{V}$ is a morphism of generalized non-symmetric ∞ -operads

$$A : \Delta_T^{\text{op}} \longrightarrow \mathcal{V}^{\otimes}.$$

Equivalently, in the algebraic-pattern language, it is a morphism of fibrous $\Delta^{\text{op}, \natural}$ -patterns from Δ_T^{op} to the generalized operad underlying $\mathcal{V}^{\otimes}$. We write

$$\text{Alg}_T(\mathcal{V}) := \text{Alg}_{\Delta_T^{\text{op}}}(\mathcal{V}^{\otimes}),$$

and $\text{Alg}_{\text{cat}}(\mathcal{V})$ for the ∞ -category in which the object anima T is allowed to vary. A map of anima $u : T \rightarrow T'$ induces a map $\Delta_T^{\text{op}} \rightarrow \Delta_{T'}^{\text{op}}$ and hence a reindexing functor

$$u^* : \text{Alg}_{T'}(\mathcal{V}) \longrightarrow \text{Alg}_T(\mathcal{V}).$$

This is the object-anima pullback used below.

This translation is conceptual rather than a change of model: Chu–Haugsgaard explicitly identify generalized non-symmetric ∞ -operads with weak Segal fibrations over $\Delta^{\text{op}, \natural}$, and Barkan–Haugsgaard–Steinebrunner identify the same objects as fibrous patterns. We continue to use the Gepner–Haugsgaard notation Δ_T^{op} because their categorical-algebra and completion theorems are stated in that notation.

The following point is crucial for the proof. By Gepner–Haugsgaard [GH15, Definition 3.1.17], $\text{Alg}_{\mathcal{M}}(\mathcal{N})$ is the *full subcategory* of the appropriate functor

∞ -category spanned by morphisms of generalized non-symmetric ∞ -operads. Thus, if $A, B \in \text{Alg}_T(\mathcal{V})$, then

$$\text{Map}_{\text{Alg}_T(\mathcal{V})}(A, B)$$

is the mapping anima between A and B in

$$\text{Fun}_{\Delta^{\text{op}}}(\Delta_T^{\text{op}}, \mathcal{V}^{\otimes}).$$

A morphism of categorical algebras is therefore an edge in this functor ∞ -category, equivalently a functor

$$H : \Delta_T^{\text{op}} \times \Delta^1 \longrightarrow \mathcal{V}^{\otimes}$$

over Δ^{op} with prescribed endpoints. In the pattern language, this is a genuine morphism in the ∞ -category of fibrous $\Delta^{\text{op}, \natural}$ -patterns, not merely objectwise data. Below, every nontrivial algebra morphism is obtained as the image of an actual morphism under a functor of ∞ -categories.

2.2. Change of base in an arbitrary monoidal base. We use Gepner–Haugsgeng’s convention that a lax monoidal functor is a morphism of non-symmetric ∞ -operads between monoidal ∞ -categories [GH15, Definition 3.1.20]. Let $\text{Mon}_{\infty}^{\text{lax}}$ denote the resulting ∞ -category. Gepner–Haugsgeng Lemma 4.3.9 states that categorical algebras are functorial in the enriching monoidal ∞ -category:

$$(1) \quad \text{Alg}_{\text{cat}}(-) : \text{Mon}_{\infty}^{\text{lax}} \longrightarrow \text{Cat}_{\infty}.$$

Thus a lax monoidal functor

$$P^{\otimes} : \mathcal{V}^{\otimes} \longrightarrow \mathcal{W}^{\otimes}$$

induces a change-of-base functor

$$(2) \quad P_* : \text{Alg}_{\text{cat}}(\mathcal{V}) \longrightarrow \text{Alg}_{\text{cat}}(\mathcal{W}),$$

which leaves the object anima unchanged [GH15, Lemma 4.3.9]. On a fixed-object algebra $A : \Delta_T^{\text{op}} \rightarrow \mathcal{V}^{\otimes}$, this is simply postcomposition,

$$P_* A = P^{\otimes} \circ A : \Delta_T^{\text{op}} \longrightarrow \mathcal{W}^{\otimes}.$$

For an ∞ -category K , write $\text{Fun}(K, \mathcal{W})_{\text{pt}}^{\otimes}$ for the pointwise monoidal functor ∞ -category. Formally, a monoidal ∞ -category is an associative algebra object in Cat_{∞} with its cartesian monoidal structure, and the right adjoint $\text{Fun}(K, -) : \text{Cat}_{\infty} \rightarrow \text{Cat}_{\infty}$ preserves products; applying it to the associative algebra object $\mathcal{W}^{\otimes}$ produces $\text{Fun}(K, \mathcal{W})_{\text{pt}}^{\otimes}$. Its underlying ∞ -category is $\text{Fun}(K, \mathcal{W})$, and evaluation at every $k \in K$ is monoidal.

Lemma 2.1 (Algebras in a pointwise monoidal functor category). *Let K be an ∞ -category, let T be an anima, and let $\mathcal{W}^{\otimes}$ be a monoidal ∞ -category. There is a canonical equivalence*

$$(3) \quad \text{Alg}_T(\text{Fun}(K, \mathcal{W})_{\text{pt}}) \simeq \text{Fun}(K, \text{Alg}_T(\mathcal{W})).$$

These equivalences are natural in K and in lax monoidal functors in $\mathcal{W}$. The symmetric-operadic analogue is Lurie [Lur17, Remark 2.1.3.4]; the associative one-object case already appears in Lurie [Lur09, Remark 1.1.18]. We include the short proof because we need the fixed-object generalized non-symmetric version.

Proof. Under straightening, the monoidal ∞ -category $\mathcal{W}^\otimes \rightarrow \Delta^{\text{op}}$ is represented by the associative-algebra diagram whose value at $[n]$ is $\mathcal{W}^n$. Applying $\text{Fun}(K, -)$ levelwise gives the diagram

$$[n] \longmapsto \text{Fun}(K, \mathcal{W}^n) \simeq \text{Fun}(K, \mathcal{W})^n,$$

and unstraightening this whole diagram is, by definition, the pointwise monoidal structure on $\text{Fun}(K, \mathcal{W})$. Equivalently, its total generalized operad is

$$(4) \quad \text{Fun}(K, \mathcal{W})_{\text{pt}}^\otimes \simeq \Delta^{\text{op}} \times_{\text{Fun}(K, \Delta^{\text{op}})} \text{Fun}(K, \mathcal{W}^\otimes),$$

where $\Delta^{\text{op}} \rightarrow \text{Fun}(K, \Delta^{\text{op}})$ is the constant-diagram functor. This description includes the active maps and the inert cocartesian lifts, not merely the fibres.

By Gepner–Haugsgeng Definition 3.1.17, $\text{Alg}_T(\mathcal{W})$ is the full subcategory of

$$\text{Fun}_{\Delta^{\text{op}}}(\Delta_T^{\text{op}}, \mathcal{W}^\otimes)$$

spanned by the functors preserving inert morphisms. Using equation (4) and currying gives a canonical equivalence

$$\text{Fun}_{\Delta^{\text{op}}}(\Delta_T^{\text{op}}, \text{Fun}(K, \mathcal{W})_{\text{pt}}^\otimes) \simeq \text{Fun}(K, \text{Fun}_{\Delta^{\text{op}}}(\Delta_T^{\text{op}}, \mathcal{W}^\otimes)).$$

Under this equivalence, preservation of inert morphisms is pointwise in K . Restricting to the corresponding full subcategories proves equation (3). Naturality follows from the naturality of straightening, unstraightening, and currying. $\square$

Definition 2.2 (Monoidal natural transformation). Let

$$P^\otimes, Q^\otimes : \mathcal{V}^\otimes \longrightarrow \mathcal{W}^\otimes$$

be lax monoidal functors. A *monoidal natural transformation*

$$\eta : P^\otimes \Longrightarrow Q^\otimes$$

is, by definition, a lax monoidal functor

$$(5) \quad \widehat{\eta}^\otimes : \mathcal{V}^\otimes \longrightarrow \text{Fun}([1], \mathcal{W})_{\text{pt}}^\otimes$$

equipped with specified identifications of its composites with the monoidal evaluation functors at 0 and 1 with $P^\otimes$ and $Q^\otimes$, respectively. The underlying functor $\mathcal{V} \rightarrow \text{Fun}([1], \mathcal{W})$ is therefore an ordinary natural transformation between the underlying functors P and Q , but the monoidal transformation is the global lax monoidal lift equation (5), not merely that underlying natural transformation.

This definition is deliberately global: the coherent monoidal naturality is the structure of the single lax monoidal functor equation (5); it is not an objectwise natural transformation supplemented by a finite list of coherence diagrams.

Corollary 2.3 (Monoidal transformations and change of base). *Let $P^\otimes, Q^\otimes : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$ be lax monoidal functors, let*

$$\eta : P^\otimes \Longrightarrow Q^\otimes$$

be a monoidal natural transformation in the sense of Definition 2.2, and fix an anima T . Write $P_{,T}$ and $Q_{*,T}$ for the restrictions of change of base to T -object categorical algebras. Then there is a canonical natural transformation*

$$\eta_{*,T} : P_{*,T} \Longrightarrow Q_{*,T}$$

of functors $\mathrm{Alg}_T(\mathcal{V}) \rightarrow \mathrm{Alg}_T(\mathcal{W})$. Its component at a T -object categorical algebra is the identity on T , and on each hom-object is induced by the underlying component of η .

Proof. Apply change of base to

$$\tilde{\eta}^\otimes : \mathcal{V}^\otimes \longrightarrow \mathrm{Fun}([1], \mathcal{W})_{\mathrm{pt}}^\otimes.$$

For fixed T this gives

$$\mathrm{Alg}_T(\mathcal{V}) \longrightarrow \mathrm{Alg}_T(\mathrm{Fun}([1], \mathcal{W})_{\mathrm{pt}}).$$

By Lemma 2.1 with $K = [1]$, the target is canonically equivalent to

$$\mathrm{Fun}([1], \mathrm{Alg}_T(\mathcal{W})).$$

The two endpoint functors are $P_{*,T}$ and $Q_{*,T}$, so the resulting functor is exactly the asserted natural transformation. $\square$

Remark 2.4 (Reference status). Corollary 2.3 is the fixed-object form of the functoriality implicit in Gepner–Haugsgeng change of base [GH15, Lemma 4.3.9]; the proof above reduces it to the pointwise-algebra equivalence Lemma 2.1. This fixed-object form is all that is used below.

Remark 2.5 (The cartesian specialization). Suppose $\mathcal{V}$ and $\mathcal{W}$ admit finite products and are equipped with their cartesian symmetric monoidal structures. Let $P, Q : \mathcal{V} \rightarrow \mathcal{W}$ preserve finite products and let $\eta : P \Rightarrow Q$ be an ordinary natural transformation. It classifies a functor

$$\tilde{\eta} : \mathcal{V} \longrightarrow \mathrm{Fun}([1], \mathcal{W}).$$

Finite products in the arrow ∞ -category are computed pointwise, so $\tilde{\eta}$ preserves finite products because its evaluations at 0 and 1 are P and Q . Lurie Corollary 2.4.1.9 identifies finite-product-preserving functors with monoidal functors between the canonical cartesian symmetric monoidal structures [Lur17, Corollary 2.4.1.9]. Applying this equivalence to $\tilde{\eta}$ gives a canonical monoidal refinement

$$\tilde{\eta}^\times : \mathcal{V}^\times \longrightarrow \mathrm{Fun}([1], \mathcal{W})_{\mathrm{pt}}^\times$$

whose endpoint evaluations are the canonical refinements $P^\times$ and $Q^\times$. Thus every ordinary natural transformation between finite-product-preserving functors determines a monoidal natural transformation in the sense of Definition 2.2.

2.3. Closed self-enrichment in an arbitrary monoidal base. A monoidal ∞ -category $\mathcal{V}^\otimes$ is *left-closed* if for every $X \in \mathcal{V}$ the functor

$$X \otimes - : \mathcal{V} \longrightarrow \mathcal{V}$$

admits a right adjoint, denoted $[X, -]_{\mathcal{V}}$. Thus there are natural equivalences

$$\mathrm{Map}_{\mathcal{V}}(X \otimes A, Y) \simeq \mathrm{Map}_{\mathcal{V}}(A, [X, Y]_{\mathcal{V}}).$$

A left action of a monoidal ∞ -category $\mathcal{U}$ on an ∞ -category $\mathcal{M}$ will be called *closed* if, for every $X \in \mathcal{M}$, the action functor $A \mapsto A \odot X$ has a right adjoint. This is the notion used by Heine and Lurie; see Heine [Hei23, Theorem 1.1 and Section 3].

We use Heine's comparison only through the following package. Let

$$\kappa^{\mathrm{H}} : \mathrm{PreCat}_{\infty}^{\mathrm{H}} \longrightarrow \mathcal{S} \times \mathrm{Op}_{\infty}$$

denote Heine's total ∞ -category of enriched precategories. Gepner–Haugsgang use the reversed enriching operad: Heine Remark 4.5 identifies a Gepner–Haugsgang $\mathcal{V}$ -categorical algebra with a Heine precategory enriched in $(\mathcal{V}^\otimes)^{\mathrm{rev}}$ [Hei23, Remark 4.5]. To work throughout in Gepner–Haugsgang coordinates, set

$$(6) \quad \mathrm{PreCat}_{\infty}^{\mathrm{GH}} := (\mathrm{id}_{\mathcal{S}} \times \mathrm{rev})^* \mathrm{PreCat}_{\infty}^{\mathrm{H}}.$$

Then the fibre over $(T, \mathcal{V}^\otimes)$ is $\mathrm{Alg}_T(\mathcal{V})$. By Heine Remark 4.41, the projection

$$\kappa^{\mathrm{GH}} : \mathrm{PreCat}_{\infty}^{\mathrm{GH}} \longrightarrow \mathcal{S} \times \mathrm{Op}_{\infty}$$

is cartesian in the object-anima variable and cocartesian in the enriching-operad variable [Hei23, Remark 4.41]. Moreover, by Heine Remark 7.11, an edge $\mathcal{A} \rightarrow \mathcal{B}$ over a map $P^\otimes : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$ and an equivalence on object animae is cocartesian exactly when

$$P(\mathcal{A}(x, y)) \longrightarrow \mathcal{B}(x, y)$$

is an equivalence for every pair of objects [Hei23, Remark 7.11].

For a left-closed $\mathcal{V}$, the regular right action $(X, A) \mapsto X \otimes A$ is equivalently a closed left action of $\mathcal{V}^{\mathrm{rev}}$ on $\mathcal{V}$. Heine's comparison functor sends this closed action, functorially in morphisms of such actions, to a Gepner–Haugsgang categorical algebra [Hei23, Theorem 4.59 and Proposition 6.10(5)]. We denote it by

$$(7) \quad \bar{\mathcal{V}} \in \mathrm{Alg}_{\mathcal{V}^\approx}(\mathcal{V})$$

and call it the *raw self-enrichment*. Its hom-object is

$$\bar{\mathcal{V}}(X, Y) \simeq [X, Y]_{\mathcal{V}}.$$

Gepner–Haugsgeng Corollary 7.4.10 gives the corresponding self-enrichment directly; we use the Heine realization because its functoriality will be needed below. No completeness assertion is included in the notation $\overline{\mathcal{V}}$ [GH15, Corollary 7.4.10 and Remark 7.4.8].

Lemma 2.6 (Composition in the raw self-enrichment). *In Gepner–Haugsgeng’s convention, the binary composition in $\overline{\mathcal{V}}$ is*

$$[X, Y]_{\mathcal{V}} \otimes [Y, Z]_{\mathcal{V}} \longrightarrow [X, Z]_{\mathcal{V}},$$

and is the mate of the iterated evaluation map

$$X \otimes ([X, Y]_{\mathcal{V}} \otimes [Y, Z]_{\mathcal{V}}) \simeq (X \otimes [X, Y]_{\mathcal{V}}) \otimes [Y, Z]_{\mathcal{V}} \longrightarrow Y \otimes [Y, Z]_{\mathcal{V}} \longrightarrow Z.$$

Proof. Heine’s convention has the reversed order of the two hom-objects, while Gepner–Haugsgeng’s convention has the order displayed above [Hei23, Remark 4.5]. For a closed action, Heine identifies the enriched hom-object with the object representing the right adjoint to the action functor [Hei23, Theorem 1.1 and Proposition 6.10(5)]. Applied to the regular action, the enriched composition is therefore characterized, under the adjunction defining $[X, -]_{\mathcal{V}}$, by two successive evaluations. This gives the displayed mate. The categorical-algebra structure, including all higher associativity and unit data, is already supplied by Heine’s comparison functor; the calculation only identifies its binary operation. $\square$

Proposition 2.7 (Closed comparison in a monoidal base). *Let $\mathcal{V}^{\otimes}$ and $\mathcal{W}^{\otimes}$ be left-closed monoidal ∞ -categories, and let*

$$P^{\otimes} : \mathcal{V}^{\otimes} \longrightarrow \mathcal{W}^{\otimes}$$

be a lax monoidal functor with underlying functor $P : \mathcal{V} \rightarrow \mathcal{W}$. Put $T = \mathcal{V}^{\simeq}$ and $p = P^{\simeq} : T \rightarrow \mathcal{W}^{\simeq}$. There is a canonical morphism

$$(8) \quad \Theta_P : P_* \overline{\mathcal{V}} \longrightarrow p^* \overline{\mathcal{W}}$$

in $\text{Alg}_T(\mathcal{W})$. Composing with the cartesian reindexing edge $p^ \overline{\mathcal{W}} \rightarrow \overline{\mathcal{W}}$ gives a morphism of varying-object categorical algebras*

$$(9) \quad \widehat{\Theta}_P : P_* \overline{\mathcal{V}} \longrightarrow \overline{\mathcal{W}}$$

whose map on object animae is p .

For $X, Y \in \mathcal{V}$, the hom-object component of equation (8) is

$$(10) \quad \theta_{X,Y} : P([X, Y]_{\mathcal{V}}) \longrightarrow [PX, PY]_{\mathcal{W}},$$

characterized as the mate of

$$(11) \quad PX \otimes P([X, Y]_{\mathcal{V}}) \longrightarrow P(X \otimes [X, Y]_{\mathcal{V}}) \xrightarrow{P(\text{ev}_{X,Y})} PY.$$

Proof. The lax monoidal structure of P gives, after reversal, a morphism from the regular closed left action of $\mathcal{V}^{\text{rev}}$ on $\mathcal{V}$ to that of $\mathcal{W}^{\text{rev}}$ on $\mathcal{W}$: in unreversed notation its action comparison is

$$PX \otimes PA \longrightarrow P(X \otimes A).$$

This is precisely a morphism of weak left actions in the sense of Heine Definition 3.20 in the present regular case [Hei23, Definition 3.20]. Applying Heine's genuine comparison functor [Hei23, Theorem 4.59] and using the coordinates equation (6) gives an edge

$$e_P : \bar{\mathcal{V}} \longrightarrow \bar{\mathcal{W}}$$

in $\text{PreCat}_\infty^{\text{GH}}$ lying over $(p, P^\otimes)$.

Use first the cartesian structure in the object-anima variable: the cartesian lift

$$p^*\bar{\mathcal{W}} \longrightarrow \bar{\mathcal{W}}$$

over (p, id) factors e_P , uniquely up to a contractible anima of choices, through an edge

$$e'_P : \bar{\mathcal{V}} \longrightarrow p^*\bar{\mathcal{W}}$$

over $(\text{id}_T, P^\otimes)$. Next take a cocartesian lift in the enriching-operad variable

$$c_P : \bar{\mathcal{V}} \longrightarrow P_! \bar{\mathcal{V}}.$$

Its universal property produces a vertical map

$$P_! \bar{\mathcal{V}} \longrightarrow p^*\bar{\mathcal{W}}.$$

The cocartesian transport $P_!$ is canonically the Gepner–Haugsgeng change of enrichment P_* . Indeed, the tautological postcomposition edge

$$\mathcal{A} \longrightarrow P_* \mathcal{A} = P^\otimes \circ \mathcal{A}$$

has unchanged object anima, and on every hom-object the comparison

$$P(\mathcal{A}(x, y)) \longrightarrow (P_* \mathcal{A})(x, y)$$

is the identity. Heine's cocartesian-edge criterion quoted above therefore says that this edge is cocartesian. Contractible uniqueness of cocartesian lifts gives $P_! \mathcal{A} \simeq P_* \mathcal{A}$. Applying this to $\bar{\mathcal{V}}$ gives equation (8); composing with the cartesian reindexing edge gives equation (9).

Finally, Heine identifies the hom-object attached to a closed action with the object representing the right adjoint to the action functor [Hei23, Theorem 1.1 and Proposition 6.10(5)]. Under this identification, the map on representing objects induced by the action comparison is the mate of equation (11). This is exactly equation (10). $\square$

Remark 2.8 (Cartesian closed categories). If $\mathcal{V}$ and $\mathcal{W}$ are cartesian closed, then they are left-closed for the cartesian tensor product. A finite-product-preserving functor $P : \mathcal{V} \rightarrow \mathcal{W}$ has a canonical strong symmetric monoidal refinement by Lurie Corollaries 2.4.1.8–2.4.1.9, and Proposition 2.7 specializes to

$$P([X, Y]_{\mathcal{V}}) \longrightarrow [PX, PY]_{\mathcal{W}},$$

whose mate is

$$P([X, Y]) \times PX \simeq P([X, Y] \times X) \longrightarrow PY.$$

This is the form used below.

2.4. Completeness. Let E^n denote the Gepner–Haugsgeng categorical algebra freely encoding a string of n composable equivalences [GH15, Definition 5.1.6]. For a categorical algebra $\mathcal{A}$ they define

$$\iota_n \mathcal{A} := \text{Map}(E^n, \mathcal{A}), \quad \iota \mathcal{A} := |\iota_\bullet \mathcal{A}|$$

[GH15, Definition 5.1.13]. The algebra $\mathcal{A}$ is *complete* when the canonical map

$$\iota_0 \mathcal{A} \longrightarrow \iota \mathcal{A}$$

is an equivalence [GH15, Definitions 5.2.1–5.2.2]; equivalently, $s_0 : \iota_0 \mathcal{A} \rightarrow \iota_1 \mathcal{A}$ is an equivalence [GH15, Corollary 5.2.10]. Thus completeness is the enriched Rezk condition: the chosen object anima agrees with the anima of objects up to equivalence.

A morphism of categorical algebras is *fully faithful* if it induces equivalences on all hom-objects, and *essentially surjective* if every target object is equivalent to one in its image; see Gepner–Haugsgeng Definitions 5.3.1 and 5.3.3 [GH15, Definitions 5.3.1 and 5.3.3]. We will not prove completeness of our raw categorical algebras directly. Instead we use their completion theorem.

Theorem 2.9 (Gepner–Haugsgeng completion). *Let $\mathcal{V}^\otimes$ be a large monoidal ∞ -category. The inclusion*

$$\text{Cat}_\infty^\mathcal{V} \hookrightarrow \text{Alg}_{\text{cat}}(\mathcal{V})$$

of complete $\mathcal{V}$ -enriched categorical algebras admits a left adjoint

$$L_\mathcal{V} : \text{Alg}_{\text{cat}}(\mathcal{V}) \longrightarrow \text{Cat}_\infty^\mathcal{V}$$

which exhibits $\text{Cat}_\infty^\mathcal{V}$ as the localization of $\text{Alg}_{\text{cat}}(\mathcal{V})$ at the fully faithful and essentially surjective enriched functors [GH15, Theorem 5.6.6]. If $\mathcal{V}$ is presentably monoidal, the explicit completion map constructed by Gepner–Haugsgeng is fully faithful and essentially surjective and has complete target [GH15, Theorem 5.6.2].

In particular, whenever the completion unit is represented by the Gepner–Haugsgeng completion map, it does not alter hom-objects between objects coming from the original categorical algebra. We apply this only to $\mathcal{V} = \text{Cat}_\infty$ with its cartesian monoidal structure, after enlarging universes so that the relevant presentability hypotheses hold.

3. THE HHLN INPUT

Definition 3.1 (Adequate triple). An *adequate triple* is a triple

$$X = (X, X_{\text{in}}, X_{\text{eg}})$$

consisting of an ∞ -category X and two wide subcategories of *ingressive* and *egressive* arrows such that, whenever $f : y \rightarrow x$ is ingressive and $g : x' \rightarrow x$

is egressive, a pullback

$$\begin{array}{ccc} y' & \xrightarrow{f'} & x' \\ g' \downarrow & & \downarrow g \\ y & \xrightarrow{f} & x \end{array}$$

exists with f' ingressive and g' egressive. A square with horizontal arrows ingressive and vertical arrows egressive is *ambigressive*; a morphism of adequate triples is a functor preserving the two wide subcategories and ambigressive pullback squares. This is HHLN Definition 2.1 [HHLN23, Definition 2.1]. We write AdTrip for the resulting ∞ -category.

For an adequate triple X , HHLN define an ∞ -category $\text{Span}(X)$ [HHLN23, Definition 2.12 and Theorem 2.13]. Concretely, it has the same objects as X ; a morphism $x \rightarrow y$ is represented by a span

$$x \xleftarrow{e} z \xrightarrow{i} y$$

with e egressive and i ingressive, and composition is by the ambigressive pullbacks of Definition 3.1. This description determines its mapping animae and composition, while HHLN's complete-Segal construction supplies the coherent ∞ -category.

If $\mathcal{C}$ admits pullbacks, then

$$\widehat{\mathcal{C}} := (\mathcal{C}, \mathcal{C}, \mathcal{C})$$

is an adequate triple. HHLN Example 2.3(2) gives a fully faithful functor

$$(12) \quad M : \text{Cat}_{\infty}^{\text{pb}} \hookrightarrow \text{AdTrip}, \quad \mathcal{C} \mapsto \widehat{\mathcal{C}},$$

whose morphisms are exactly the pullback-preserving functors [HHLN23, Example 2.3(2)].

Proposition 3.2 (HHLN, Lemma 2.5). *The ∞ -category AdTrip is cartesian closed. Its internal hom is the adequate triple*

$$[X, Y]_{\text{AdTrip}} = \text{Fun}_{\text{AdTrip}}(X, Y).$$

The underlying ∞ -category is the full subcategory of $\text{Fun}(X, Y)$ spanned by morphisms of adequate triples. A natural transformation $\tau : F \Rightarrow G$ is ingressive precisely when

- (1) *every component τ_x is ingressive in Y , and*
- (2) *for every egressive arrow $f : x \rightarrow y$ of X , the naturality square of τ at f is cartesian.*

The evaluation map

$$[X, Y]_{\text{AdTrip}} \times X \longrightarrow Y$$

is a morphism of adequate triples and exhibits this object as the internal hom.

HHLN also prove that AdTrip has all limits, computed component-wise [HHLN23, Lemma 2.4].

Define

$$I : \text{AdTrip} \longrightarrow \text{Cat}_\infty, \quad I(X) = X_{\text{in}}.$$

Componentwise products in AdTrip imply that I preserves finite products.

The construction $X \mapsto \text{Span}(X)$ is functorial in morphisms of adequate triples. We write

$$S := \text{Span} : \text{AdTrip} \longrightarrow \text{Cat}_\infty.$$

HHLN Theorem 2.18 identifies S as a right adjoint; hence S preserves all limits, in particular finite products [HHLN23, Theorem 2.18].

Finally, HHLN Example 2.23 gives, for every adequate triple X , the forward-map inclusion

$$(13) \quad j_X : X_{\text{in}} \longrightarrow \text{Span}(X), \quad f : x \rightarrow y \longmapsto \left(x \xleftarrow{\text{id}} x \xrightarrow{f} y \right).$$

Lemma 3.3 (Naturality of the forward-map inclusion). *The functors equation (13) assemble to a natural transformation*

$$(14) \quad j : I \Longrightarrow S$$

of functors $\text{AdTrip} \rightarrow \text{Cat}_\infty$.

Proof. HHLN Corollary 2.22 constructs the comparison from forward maps to spans naturally in the adequate triple X , and Example 2.23 identifies its value on X with the functor equation (13) [HHLN23, Corollary 2.22 and Example 2.23]. Therefore the j_X assemble to the natural transformation equation (14). $\square$

4. THE RAW SOURCE CATEGORICAL ALGEBRA

Let

$$\overline{\text{AdTrip}} \in \text{Alg}_{\text{AdTrip}^\simeq}(\text{AdTrip})$$

be the raw self-enrichment of equation (7); its hom-object from X to Y is the HHLN internal hom $[X, Y]_{\text{AdTrip}}$. Change base along I :

$$I_* \overline{\text{AdTrip}} \in \text{Alg}_{\text{AdTrip}^\simeq}(\text{Cat}_\infty).$$

Its hom-object from X to Y is

$$(15) \quad I([X, Y]_{\text{AdTrip}}) = [X, Y]_{\text{AdTrip}, \text{in}}.$$

Let

$$T := \text{Cat}_\infty^{\text{pb}^\simeq} \longrightarrow \text{AdTrip}^\simeq$$

be induced by the maximal-triple functor M . Precomposition with $\Delta_T^{\text{op}} \rightarrow \Delta_{\text{AdTrip}^\simeq}^{\text{op}}$ restricts the object anima to maximal triples. Define the raw source categorical algebra

$$(16) \quad \mathcal{P}_0 := T^*(I_* \overline{\text{AdTrip}}) \in \text{Alg}_T(\text{Cat}_\infty).$$

Thus $\mathcal{P}_0$ is obtained by restriction of the object anima to maximal adequate triples.

Proposition 4.1. *For $\mathcal{C}, \mathcal{D} \in \mathbf{Cat}_\infty^{\text{pb}}$ there is a canonical equivalence*

$$\mathcal{P}_0(\mathcal{C}, \mathcal{D}) \simeq \mathbf{Fun}_{\text{pb}}^{\text{cart}}(\mathcal{C}, \mathcal{D}).$$

Proof. By HHLN Example 2.3(2), the objects of

$$[\widehat{\mathcal{C}}, \widehat{\mathcal{D}}]_{\text{AdTrip}}$$

are exactly the pullback-preserving functors $\mathcal{C} \rightarrow \mathcal{D}$. By Proposition 3.2, an ingressive morphism $\alpha : F \Rightarrow G$ in this internal hom is a natural transformation that is pointwise ingressive in $\widehat{\mathcal{D}}$ and cartesian on every egressive arrow of $\widehat{\mathcal{C}}$. In a maximal triple every arrow is both ingressive and egressive. Hence the pointwise condition is vacuous and the second condition says exactly that every naturality square of α is a pullback. Using equation (15) gives the claimed equivalence of ∞ -categories. $\square$

Proposition 4.2 (Composition in the raw source). *Under the equivalences of Proposition 4.1, the enriched composition of $\mathcal{P}_0$ is the ordinary composition functor*

$$\mathbf{Fun}_{\text{pb}}^{\text{cart}}(\mathcal{C}, \mathcal{D}) \times \mathbf{Fun}_{\text{pb}}^{\text{cart}}(\mathcal{D}, \mathcal{E}) \longrightarrow \mathbf{Fun}_{\text{pb}}^{\text{cart}}(\mathcal{C}, \mathcal{E}), \quad (F, G) \longmapsto G \circ F.$$

In particular, its action on morphisms is the usual whiskering and vertical composition of natural transformations.

Proof. The categorical algebra $\mathcal{P}_0$ is obtained from the categorical self-enrichment of AdTrip by change of base along $I = (-)_{\text{in}}$ and restriction of the object anima. Hence its composition is the image under I of the internal composition map

$$[X, Y]_{\text{AdTrip}} \times [Y, Z]_{\text{AdTrip}} \longrightarrow [X, Z]_{\text{AdTrip}}.$$

By Lemma 2.6, this internal composition is the mate of iterated evaluation. HHLN Lemma 2.5 identifies the underlying ∞ -category of each internal hom with the corresponding functor ∞ -category and its evaluation with ordinary evaluation [HHLN23, Lemma 2.5]. Hence the displayed composition is ordinary composition of morphisms of adequate triples, including the usual whiskering and composition on natural transformations. Applying I restricts to ingressive natural transformations. On maximal triples, Proposition 4.1 identifies these exactly with cartesian natural transformations. Thus the resulting composition is the asserted ordinary composition on $\mathbf{Fun}_{\text{pb}}^{\text{cart}}$. $\square$

We now complete rather than attempt a direct completeness argument.

Definition 4.3. Define

$$\mathbf{Cat}_\infty^{\text{pb, cart}} := L_{\mathbf{Cat}_\infty}(\mathcal{P}_0).$$

By Theorem 2.9, the completion map

$$u_{\mathcal{P}} : \mathcal{P}_0 \longrightarrow \mathbf{Cat}_\infty^{\text{pb, cart}}$$

is fully faithful and essentially surjective. Therefore every object of the completed source is equivalent to one represented by some $\mathcal{C} \in \mathbf{Cat}_\infty^{\text{pb}}$, and for

represented objects Proposition 4.1 remains the exact mapping ∞ -category. Since $u_{\mathcal{P}}$ is an enriched functor and is fully faithful, the composition on represented mapping ∞ -categories is the composition identified in Proposition 4.2.

5. THE RAW SPAN MORPHISM

Change base the raw self-enrichment of AdTrip along S :

$$S_* \overline{\text{AdTrip}} \in \text{Alg}_{\text{AdTrip}^\simeq}(\text{Cat}_\infty),$$

with hom-object

$$\text{Span}([X, Y]_{\text{AdTrip}}).$$

The finite-product-preserving functors $I, S : \text{AdTrip} \rightarrow \text{Cat}_\infty$ have canonical cartesian symmetric monoidal refinements. By Lemma 3.3 and Remark 2.5, the natural transformation $j : I \Rightarrow S$ determines a monoidal natural transformation between these refinements. Applying Corollary 2.3 with $T = \text{AdTrip}^\simeq$, we obtain a morphism of Cat_∞ -categorical algebras

$$(17) \quad J : I_* \overline{\text{AdTrip}} \longrightarrow S_* \overline{\text{AdTrip}},$$

identity on the object anima, whose hom map is

$$[X, Y]_{\text{AdTrip}, \text{in}} \xrightarrow{j_{[X, Y]}} \text{Span}([X, Y]_{\text{AdTrip}}).$$

The morphism J is the component at $\overline{\text{AdTrip}}$ of the natural transformation on change-of-base functors supplied by Corollary 2.3.

Next apply Proposition 2.7, in the cartesian specialization of Remark 2.8, to

$$P = S : \text{AdTrip} \longrightarrow \text{Cat}_\infty.$$

Since both categories are cartesian closed and S preserves finite products, we obtain the varying-object morphism of equation (9),

$$(18) \quad K := \widehat{\Theta}_S : S_* \overline{\text{AdTrip}} \longrightarrow \overline{\text{Cat}_\infty}$$

with object map $X \mapsto \text{Span}(X)$ and hom map

$$(19) \quad \text{Span}([X, Y]_{\text{AdTrip}}) \longrightarrow \text{Fun}(\text{Span}(X), \text{Span}(Y)),$$

the mate of

$$\text{Span}([X, Y]) \times \text{Span}(X) \simeq \text{Span}([X, Y] \times X) \xrightarrow{\text{Span}(\text{ev})} \text{Span}(Y).$$

Restricting the composite $K \circ J$ along the object-anima map $T \rightarrow \text{AdTrip}^\simeq$ gives a morphism of categorical algebras

$$(20) \quad \Phi_0 : \mathcal{P}_0 \longrightarrow \overline{\text{Cat}_\infty}.$$

No completeness is used in constructing equation (20).

6. PASSAGE TO COMPLETE ENRICHED CATEGORIES

Define the target $(\infty, 2)$ -category by

$$(21) \quad \mathbf{Cat}_\infty := L_{\mathbf{Cat}_\infty}(\overline{\mathbf{Cat}_\infty}).$$

Let

$$u_Q : \overline{\mathbf{Cat}_\infty} \longrightarrow \mathbf{Cat}_\infty$$

be its completion map. It is fully faithful and essentially surjective by Theorem 2.9; in particular, between represented objects A, B its mapping ∞ -category is canonically equivalent to $\mathbf{Fun}(A, B)$.

Proposition 6.1 (Represented part of the target). *For represented ∞ -categories A, B, C , the mapping ∞ -categories in $\mathbf{Cat}_\infty$ are*

$$\mathbf{Cat}_\infty(A, B) \simeq \mathbf{Fun}(A, B),$$

and under these equivalences the enriched composition

$$\mathbf{Cat}_\infty(A, B) \times \mathbf{Cat}_\infty(B, C) \longrightarrow \mathbf{Cat}_\infty(A, C)$$

is ordinary composition of functors and natural transformations.

Proof. Before completion, $\overline{\mathbf{Cat}_\infty}$ is the categorical self-enrichment of the cartesian closed ∞ -category $\mathbf{Cat}_\infty$, so its hom-object from A to B is the internal hom $\mathbf{Fun}(A, B)$. By Lemma 2.6, its composition is the mate of iterated evaluation; in the cartesian closed category $\mathbf{Cat}_\infty$ this is precisely ordinary composition of functors, with ordinary whiskering and composition on natural transformations. The completion map u_Q is a fully faithful enriched functor by Theorem 2.9. Hence it identifies these hom-objects and their composition maps with those between the corresponding represented objects in the completion. $\square$

The composite

$$u_Q \circ \Phi_0 : \mathcal{P}_0 \longrightarrow \mathbf{Cat}_\infty$$

has complete target. Since

$$L_{\mathbf{Cat}_\infty} : \mathbf{Alg}_{\mathbf{cat}}(\mathbf{Cat}_\infty) \longrightarrow \mathbf{Cat}_\infty^{\mathbf{Cat}_\infty}$$

is left adjoint to the inclusion of complete objects, the adjunction gives, up to a contractible anima of choices, a factorization

$$(22) \quad \begin{array}{ccc} \mathcal{P}_0 & \xrightarrow{\Phi_0} & \overline{\mathbf{Cat}_\infty} \\ u_P \downarrow & & \downarrow u_Q \\ \mathbf{Cat}_\infty^{\text{pb, cart}} & \xrightarrow{\mathbf{Span}} & \mathbf{Cat}_\infty. \end{array}$$

Equivalently,

$$\mathbf{Map}_{\mathbf{Alg}_{\mathbf{cat}}(\mathbf{Cat}_\infty)}(L_{\mathbf{Cat}_\infty} \mathcal{P}_0, \mathbf{Cat}_\infty) \simeq \mathbf{Map}_{\mathbf{Alg}_{\mathbf{cat}}(\mathbf{Cat}_\infty)}(\mathcal{P}_0, \mathbf{Cat}_\infty)$$

and $\mathbf{Span}$ is the point corresponding to $u_Q \Phi_0$.

Because both vertical maps in equation (22) are fully faithful, for represented $\mathcal{C}, \mathcal{D}$ the map on hom ∞ -categories induced by $\mathbf{Span}$ is identified,

through canonical equivalences, with the raw hom map induced by equation (20). Thus the required map is a morphism in the ∞ -category of complete Cat_∞ -enriched categorical algebras: it is obtained from the morphism Φ_0 in $\text{Alg}_{\text{cat}}(\text{Cat}_\infty)$ by the localization adjunction.

7. IDENTIFICATION ON REPRESENTED CELLS

We identify the raw map equation (20), hence also the completed map, on objects, 1-morphisms, and 2-morphisms.

Proposition 7.1. *Let $\mathcal{C}, \mathcal{D}$ admit pullbacks.*

- (1) *The represented object $\mathcal{C}$ is sent to $\text{Span}(\mathcal{C})$.*
- (2) *A pullback-preserving functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is sent to the usual functor*

$$\text{Span}(F) : \text{Span}(\mathcal{C}) \rightarrow \text{Span}(\mathcal{D}).$$

- (3) *If $\alpha : F \Rightarrow G$ is cartesian, then at $x \in \mathcal{C}$ the component of the induced natural transformation is the forward span*

$$F(x) \xleftarrow{\text{id}} F(x) \xrightarrow{\alpha_x} G(x).$$

Proof. The object statement follows from the object map of K .

A pullback-preserving functor F is an object of the internal hom

$$[\widehat{\mathcal{C}}, \widehat{\mathcal{D}}]_{\text{AdTrip}}.$$

Under J it is the corresponding object of the span category of that internal hom. The morphism K is the closed comparison associated to the evaluation map

$$[\widehat{\mathcal{C}}, \widehat{\mathcal{D}}]_{\text{AdTrip}} \times \widehat{\mathcal{C}} \longrightarrow \widehat{\mathcal{D}}.$$

Restricting evaluation at the point F gives the morphism of adequate triples F itself; applying Span therefore gives the usual $\text{Span}(F)$.

If $\alpha : F \Rightarrow G$ is cartesian, Proposition 4.1 identifies it with an ingressive edge of the HHLN internal hom. The map J sends it, by equation (13), to the forward span

$$F \xleftarrow{\text{id}} F \xrightarrow{\alpha} G.$$

Evaluation at x gives

$$F(x) \xleftarrow{\text{id}} F(x) \xrightarrow{\alpha_x} G(x).$$

For completeness, take a span

$$x \xleftarrow{p} z \xrightarrow{q} y$$

in $\mathcal{C}$. The comparison between the two composites in the naturality square has, on apices, the canonical map

$$F(z) \longrightarrow F(x) \times_{G(x)} G(z), \quad u \longmapsto (F(p)(u), \alpha_z(u)).$$

This is an equivalence because

$$\begin{array}{ccc} F(z) & \xrightarrow{F(p)} & F(x) \\ \alpha_z \downarrow & & \downarrow \alpha_x \\ G(z) & \xrightarrow{G(p)} & G(x) \end{array}$$

is a pullback square. The right legs to $G(y)$ agree by naturality along q . Thus the induced transformation is exactly the usual $\text{Span}(\alpha)$, as in Harington–Mimram Proposition 3.7 [HM24, Proposition 3.7]. $\square$

Proof of Theorem 1.1, enriched part. Definition 4.3 and Propositions 4.1 and 4.2 and the fully faithful and essentially surjective completion map give the first two assertions about the source, including the expected composition on the displayed mapping ∞ -categories. The morphism equation (22) is a morphism between complete Cat_∞ -enriched categorical algebras, hence a Cat_∞ -enriched functor between the corresponding $(\infty, 2)$ -categories. Proposition 7.1 identifies its restriction to represented 0-, 1-, and 2-cells with the usual span construction. $\square$

8. LIMIT PRESERVATION

Proposition 8.1. *The ordinary ∞ -functor*

$$\text{Cat}_\infty^{\text{pb}} \longrightarrow \text{Cat}_\infty, \quad \mathcal{C} \longmapsto \text{Span}(\mathcal{C}),$$

preserves all small limits.

Proof. HHLN Lemma 2.4 computes limits in AdTrip componentwise. Hence the essential image of the maximal-triple embedding equation (12) is closed under limits: the limit of a diagram of maximal triples is

$$\left(\lim_i \mathcal{C}_i, \lim_i \mathcal{C}_i, \lim_i \mathcal{C}_i \right).$$

Thus $M : \text{Cat}_\infty^{\text{pb}} \rightarrow \text{AdTrip}$ creates and preserves small limits. By HHLN Theorem 2.18, $\text{Span} : \text{AdTrip} \rightarrow \text{Cat}_\infty$ is a right adjoint and therefore preserves small limits. The composite $\text{Span} \circ M$ is the usual span functor on ∞ -categories with pullbacks. $\square$

Completion of the proof of Theorem 1.1. Proposition 8.1 is precisely the final assertion. $\square$

Remark 8.2. Proposition 8.1 is a statement about the underlying $(\infty, 1)$ -functor. Harington–Mimram phrase Conjecture 3.5 as a “limit-preserving $(\infty, 2)$ -functor” [HM24, Conjecture 3.5]. If that phrase is intended to assert preservation of genuinely $(\infty, 2)$ -categorical weighted limits, that stronger statement is not proved here. What is proved here is the existence of the coherent $(\infty, 2)$ -functor together with preservation of all ordinary small limits by its object/1-morphism truncation.

9. REFERENCE AND DEPENDENCY AUDIT

For clarity, we distinguish statements quoted directly from the literature from formal consequences proved in this note.

- (1) The algebraic-pattern interpretation in Section 2.1 is standard: algebraic patterns are from Chu–Haugsgeng [CH21, Definition 2.1]; the two patterns on Δ^{op} are in Chu–Haugsgeng [CH21, Example 3.3] and Barkan–Haugsgeng–Steinebrunner [BHS25, Example 3.1.4]; generalized non-symmetric ∞ -operads are identified with weak Segal $\Delta^{\text{op}, \natural}$ -fibrations in Chu–Haugsgeng [CH21, Example 9.8(ii)] and with fibrous $\Delta^{\text{op}, \natural}$ -patterns in Barkan–Haugsgeng–Steinebrunner [BHS25, Example 4.1.9].
- (2) The span-specific results are quoted from HHLN: Example 2.3(2), Lemmas 2.4 and 2.5, Theorem 2.18, Corollary 2.22, and Example 2.23. No span-specific higher-coherence assertion is inserted beyond these references.
- (3) Change of base for categorical algebras is exactly Gepner–Haugsgeng [GH15, Lemma 4.3.9]. The pointwise-algebra equivalence Lemma 2.1 is the generalized non-symmetric fixed-object version of the standard pointwise-algebra formula; compare Lurie [Lur17, Remark 2.1.3.4] and [Lur09, Remark 1.1.18]. The proof records explicitly the total-operad pullback formula equation (4), so the passage from pointwise tensor products to algebras is made in the ambient functor ∞ -categories rather than inferred only from fibres.
- (4) The monoidal-transformation statement Corollary 2.3 is the fixed-object consequence of Gepner–Haugsgeng change of base [GH15, Lemma 4.3.9] and Lemma 2.1.
- (5) Raw self-enrichment and the closed comparison use Heine’s comparison functor [Hei23, Theorem 4.59 and Proposition 6.10(5)]. The reparametrization equation (6) records the reversal in Heine Remark 4.5. Cartesian reindexing in the object anima and cocartesian change of enrichment are supplied by Heine Remark 4.41, while the homwise criterion for cocartesian edges is Remark 7.11. These facts identify the cocartesian transport with Gepner–Haugsgeng change of base without any assumption that Heine’s comparison functor preserves cocartesian arrows.
- (6) Completion is quoted directly from Gepner–Haugsgeng [GH15, Theorems 5.6.2 and 5.6.6].
- (7) Proposition 3.2 is HHLN Lemma 2.5. Proposition 4.1 is its specialization to maximal adequate triples, while Proposition 4.2 identifies the inherited self-enrichment composition with ordinary functor composition. We do not know separate references for these short specializations.
- (8) Lemma 3.3 is extracted directly from the naturality in X of HHLN Corollary 2.22 together with Example 2.23. Proposition 7.1 is a

formal identification of the already constructed enriched functor on represented cells. Its 2-cell formula agrees with Harington–Mimram [HM24, Proposition 3.7].

- (9) Proposition 8.1 is the direct composite of HHLN Lemma 2.4 (componentwise limits in $\mathbf{AdTrip}$) and HHLN Theorem 2.18 (the span functor is a right adjoint); we have not found this restriction to maximal triples stated separately.

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